

trigonometry and geometry

Straight Angles

The sum of angles that form a straight angle is 180° .
 $\angle a + \angle b = 180^\circ$

• these are a linear pair $\angle a + \angle b = 180$

EXAMPLE

Determine the value of the unknown angle.

Solution

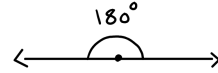
Since $\angle x$ and 77° form a straight angle, their sum is 180° .

$$77^\circ + \angle x = 180^\circ$$

$$\angle x = 180^\circ - 77^\circ$$

$$\angle x = 103^\circ$$

- a Straight line is 180 degrees



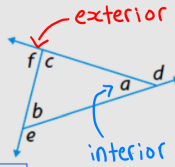
- When you are given a Problem where you have to find the missing angle you can set it equal to 180 degrees and solve for the missing angle

Interior and Exterior Angles of a Triangle

The sum of the interior angles in a triangle is 180° .

$$\angle a + \angle b + \angle c = 180^\circ$$

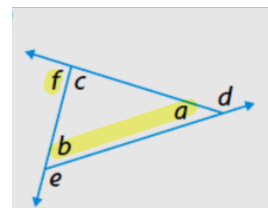
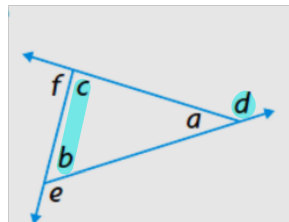
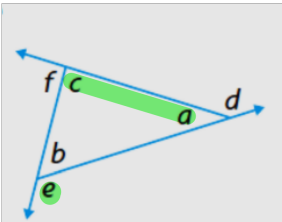
Each exterior angle equals the sum of the two interior angles opposite it.



- anything outside the triangle is called exterior angle anything inside the triangle is called interior angle

$\angle d = \angle b + \angle c$	$\angle e = \angle a + \angle c$	$\angle f = \angle a + \angle b$
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- when combining the interior together they add up to 180 degrees



- Notice that right in front each angle is what they are equal to

EXAMPLE

Determine the value for the unknown angle.

Solution

The sum of the interior angles in a triangle is 180° .

$$\angle x + 73^\circ + 69^\circ = 180^\circ$$

$$\angle x = 180^\circ - 69^\circ - 73^\circ$$

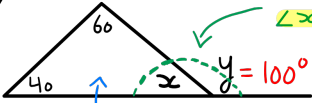
$$\angle x = 38^\circ$$



- inside the triangle the angle all add up to 180 degrees

example Problems

#1



these are a linear pair
they add up to 180°

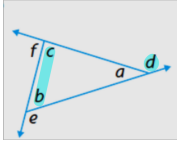
$$\angle x + \angle y = 180$$

inside the triangle
all the angles add
up to 180°

$$40 + 60 + x = 180$$

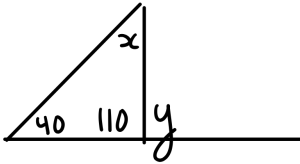
$$100 + x = 180$$

$$x = 80$$



$$y = 40^\circ + 60^\circ$$
$$y = 100^\circ$$

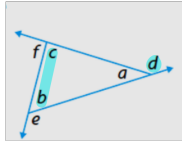
#2



$$40 + 110 + x = 180$$

$$150 + x = 180$$

$$x = 30$$

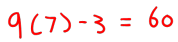


$$y = 40 + x \quad x = 30$$

$$y = 40 + 30$$

$$y = 70$$

#3 Find the missing angle



$$6(7) + 8 = 50 \quad 60 + 50 + ? = 180$$

$$110 + ? = 180$$

$$110 + ? = 180$$

$$10x + 40 = 15x + 5$$

$$x = 7$$

Angle Properties of Parallel Lines

When a transversal crosses 2 parallel lines:

- Corresponding angles are equal.

$$\angle a = \angle e$$

$$\angle c = \angle g$$

$$\angle b = \angle f$$

$$\angle d = \angle h$$

- Alternate angles are equal.

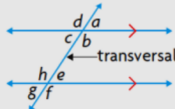
$$\angle b = \angle d$$

$$\angle c = \angle e$$

- The sum of the interior angles on the same side of the transversal is 180° .

$$\angle b + \angle e = 180^\circ$$

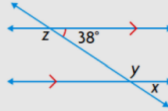
$$\angle c + \angle f = 180^\circ$$



EXAMPLE

Determine the values of the unknown angles.

Explain your solution.



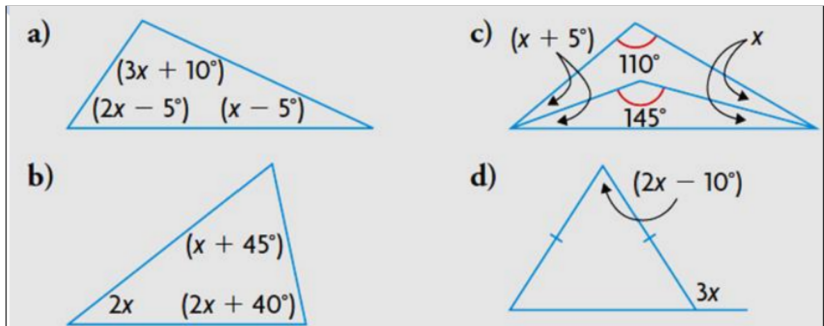
Solution

- $\angle x = 38^\circ$ since the angles are corresponding angles.
- The sum of the interior angles on the same side of the transversal is 180° . This means that $\angle y = 180^\circ - 38^\circ = 142^\circ$.
- $\angle z = \angle y = 142^\circ$ since the two angles are alternate angles.

- the sum of interior angles on that are on the same side of the transversal are equal to 180 degrees

$$\angle 38^\circ + \angle y = 180$$

$$y = 180 - 38 \Rightarrow y = 142^\circ$$



- a) angles in a triangle are equal to 180 degrees

$$(3x + 10) + (2x - 5) + (x - 5) = 180$$

$$6x = 180$$

$$x = 30$$

- b) angles add up to 180 in a triangle

$$(2x) + (x + 45) + (2x + 40) = 180$$

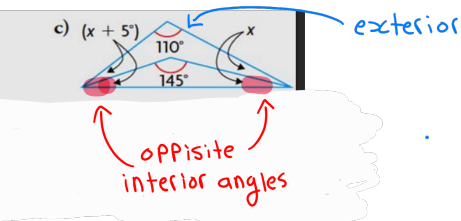
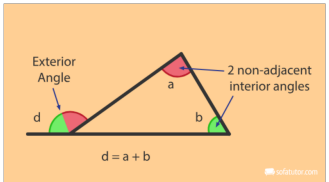
$$5x + 85 = 180$$

$$5x = 95$$

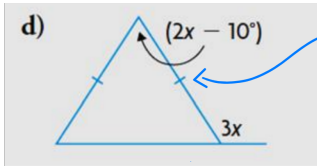
$$x = 19$$

©

- 110° is the exterior angle, so the exterior angle equals the sum of the two opposite interior angles



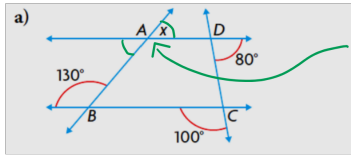
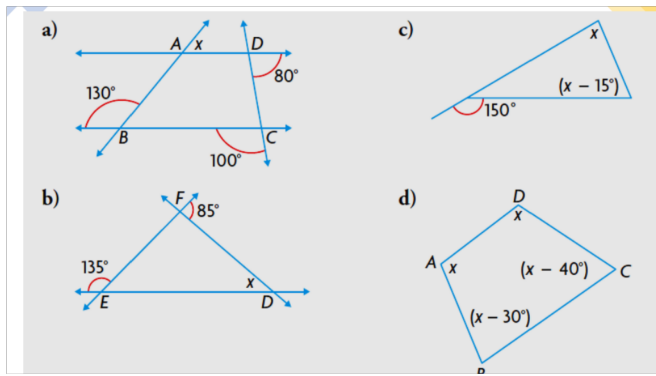
$$\begin{aligned}(x) + (x + 5) &= 110 \\ 2x + 5 &= 110 \\ 2x &= 105 \\ \boxed{x} &= 52.5\end{aligned}$$



- those two tick mark tell you they are equal to each other

- this means both sides with tick marks angles are equal to each other
- also a triangles angle add up to 180°

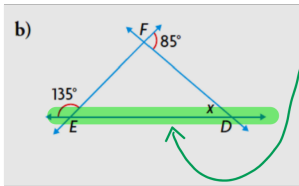
$$\begin{aligned}(2x - 10) + 3x + 3x &= 180 \\ 8x - 10 &= 180 \\ 8x &= 190 \\ \boxed{x} &= 23.7\end{aligned}$$



$$130 + x = 180$$

$$x = 50$$

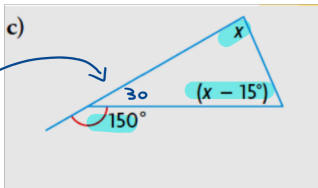
- We have vertical angles which are equal to other so now they are both on the same side of the transversal so both angle added together must equal 180



$$135 + x = 180$$

$$x = 45$$

- a straight line is 180° we already know that there is an angle that is 135° and there is another angle "x" we need to solve for



$$150 = x + (x - 15)$$

$$150 = 2x - 15$$

$$165 = 2x$$

$$x = 82.5$$

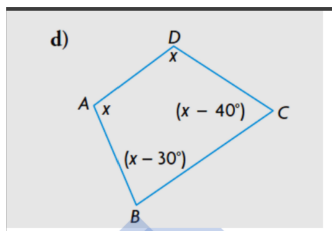
- the exterior angle is equal to sum of the two opposite interior angles

- We can check our answer by plugging $x = 82.5$ and seeing if we get 180° for the triangle

$$30 + (82.5) + (82.5 - 15)$$

$$30 + 82.5 + 67.5 = 180$$

straight line which is 180°
 $180 - 150 = 30$



- this is a quadrilateral
its just two triangles
each triangle is 180°
they total 360°

$$(x) + (x) + (x - 40) + (x - 30) = 360$$

$$4x - 70 = 360$$

$$4x = 430$$

$$x = 107.5$$

Congruent vs Similar triangle

◆ Congruent Triangles ($\cong$)

- Same shape
- Same size
- All corresponding **angles** are equal.
- All corresponding **sides** are equal.
- Example: Two identical triangle cutouts.

◆ Similar Triangles ($\sim$)

- Same shape
- Different size
- All corresponding **angles** are equal.
- Corresponding **sides are in the same ratio** (not necessarily equal).
- Example: One triangle is a **scaled-up** or **scaled-down** version of another.

Similar triangles

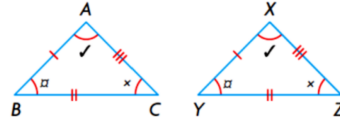
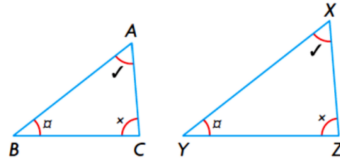
Need to Know

- If $\angle A = \angle X$, $\angle B = \angle Y$, and $\angle C = \angle Z$, then

$$\triangle ABC \sim \triangle XYZ \text{ and } \frac{AB}{XY} = \frac{BC}{YZ} = \frac{AC}{XZ}$$

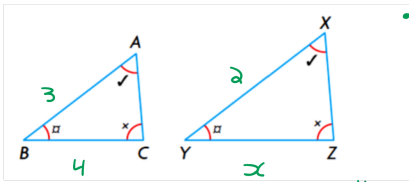
- If $\angle A = \angle X$, $\angle B = \angle Y$, and $\angle C = \angle Z$, and if $AB = XY$ or $BC = YZ$ or $AC = XZ$, then $\triangle ABC \cong \triangle XYZ$.

- When comparing similar triangles, if the scale factor is
 - greater than 1, the larger triangle is an enlargement of the smaller triangle
 - between 0 and 1, the smaller triangle is a reduction of the larger triangle
 - 1, the triangles are congruent



Congruent

- this states if angle a is equal to angle x and angle angle b is equal to angle y and angle c is equal to angle z then these are similar triangles ($\triangle ABC \sim \triangle xyz$)



$$\frac{AB}{xy} = \frac{BC}{yz} \Rightarrow \frac{3}{2} = \frac{4}{x}$$

$$3x = 8$$

$$x = 2.6$$

- if you wanted to solve for x you can use this equation $\frac{AB}{xy} = \frac{BC}{yz} = \frac{AC}{xz}$ you can only this equation when ($\triangle ABC \sim \triangle xyz$)

- you can solve for a missing side even without knowing all three sides

when you are writing the equation you will divide equal sides

- as soon as you know the triangles are similar you're allowed to use any matching pair of sides to solve

Scale Factor

#1

$$\text{scale factor} = \frac{\text{larger length}}{\text{smaller length}}$$

#2

$$\text{scale factor} = \frac{\text{smaller length}}{\text{larger length}}$$

- the scale factor is a number used to figure out if a figure has increased or decreased

#1 When you are going from small to large you are scaling up

#2 When you are going from large to small you are scaling down

- the answer you get tells you if it's just an enlargement of the smaller triangle if it's a reduction of the larger triangle or if they are congruent

Example:

Show that the two triangles in this diagram are similar. Then determine the values of x and y .

these are similar triangles

$$\begin{aligned}\angle BAD &= \angle CED \\ \angle ABD &= \angle ECD \\ \text{So, } \triangle ABD &\sim \triangle ECD.\end{aligned}$$

$$\begin{aligned}\frac{BD}{CD} &= \frac{AB}{EC} \\ \frac{14.0}{6.0} &= \frac{x}{6.0} \\ 6.0 \left(\frac{14.0}{6.0} \right) &= 6.0 \left(\frac{x}{6.0} \right) \\ 14.0 &= x\end{aligned}$$

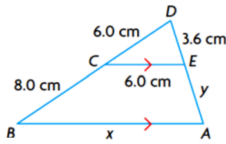
$$\frac{AD}{ED} = \frac{BD}{CD}$$

$$\frac{y + 3.6}{3.6} = \frac{14.0}{6.0}$$

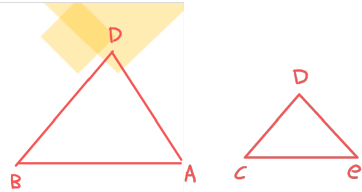
Since AB and EC are parallel, the corresponding angles in the small and large triangles are equal.

To calculate x , I set up a proportion using the corresponding sides I knew and the side I needed to determine in the similar triangles. Then I solved for x .

To calculate y , I set up another proportion using the corresponding sides I knew and a side that contains y in the similar triangles. Then I solved for y .



$\Rightarrow$



$$\frac{BD}{CD} = \frac{BA}{CE} \quad \text{or} \quad \frac{BD}{CD} = \frac{AB}{EC}$$

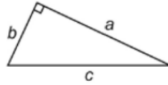
Same thing

$$\begin{aligned}3.6 \left(\frac{y + 3.6}{3.6} \right) &= 3.6 \left(\frac{7.0}{3.0} \right) \\ y + 3.6 &= 8.4 \\ y &= 4.8\end{aligned}$$

The value of x is 14.0 cm, and the value of y is 4.8 cm.

The Pythagorean Theorem

Main
Formula



$$a^2 + b^2 = c^2$$

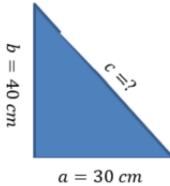
Remember

$$c^2 = a^2 + b^2$$

$$a^2 = c^2 - b^2$$

$$b^2 = c^2 - a^2$$

Example: Find the hypotenuse of a triangle with one side 30cm and other side 40cm.



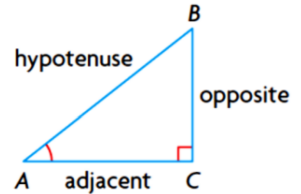
$$\begin{aligned} c^2 &= a^2 + b^2 \\ c^2 &= 30^2 + 40^2 \\ c^2 &= 900 + 1600 \\ c^2 &= 2500 \\ \sqrt{c^2} &= \sqrt{2500} \\ c &= 50 \text{ cm} \end{aligned}$$

The Primary Trigonometric Ratios

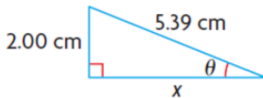
• remember SOH-CAH-TOA

- If $\angle A$ is one of the acute angles in a right triangle, the three primary trigonometric ratios for $\angle A$ can be written as

$$\begin{aligned} \sin A &= \frac{\text{opposite}}{\text{hypotenuse}} \\ \cos A &= \frac{\text{adjacent}}{\text{hypotenuse}} \\ \tan A &= \frac{\text{opposite}}{\text{adjacent}} \end{aligned}$$



- Using the Pythagorean theorem, $\text{opposite}^2 + \text{adjacent}^2 = \text{hypotenuse}^2$ in any right triangle.



$$\begin{aligned} \sin \theta &= \frac{\text{opposite}}{\text{hypotenuse}} \\ \sin \theta &= \frac{2.00}{5.39} \end{aligned}$$

$$\theta = \sin^{-1}\left(\frac{2.00}{5.39}\right)$$

$$\theta \doteq 21.8^\circ$$

$$\theta \doteq 22^\circ$$

$$\begin{aligned} x^2 + 2.00^2 &= 5.39^2 \\ x^2 + 4.00 &= 29.0521 \\ x^2 &= 29.0521 - 4.00 \\ x^2 &= 25.0521 \\ x &= \sqrt{25.0521} \\ x &\doteq 5.01 \end{aligned}$$

I started by determining θ using the sine ratio. If I knew the angle, its sine would be $\frac{2.00}{5.39}$.

Since I knew the sine and not the angle, I had to use the **inverse** of sine, which is $\sin^{-1}$.

Since the tangent and cosine ratios both involve the adjacent side, x , I calculated the length of this side using the Pythagorean theorem.